

Master 1 Innovation, Entreprise et Société (IES)
Cursus Master en Ingénierie (CMI) - Data Science for Social Sciences (D3S)



Investigating the changes in Eugene's response to non-violent crisis emergencies

Internship Report

Academic Year 2025-2026

Author:

Mathilde LAFAY

Student ID: 42002854

Academic Supervisor (UO):

Dr. Rori ROHLFS

Teaching Tutor (UPN):

Dr. Sana BEN HAMIDA

June 23, 2026

Acknowledgements

I would like to thank Dr. Rori Rohlf for her guidance, support and the opportunity to work at the Rohlfslab. I would also like to extend my gratitude to Nathan Burton and the Data Science in Action class of 2026 for their advice and for being a great source of inspiration and reflection throughout this whole project, as well as the other Rohlfslab members, Carolina, Sebastián, Myco, Melody, Alicia, and Gisselle for making this experience so positive.

I also thank the members of the Global Studies Institute of the University of Oregon, Elly Vandegrift and Meaghan Boghart, as well as Laurel Rosch, the Big Ten Academic Alliance IREP coordinator and the FIGURE Network for creating and making this research opportunity abroad possible.

Lastly, I would like to express my gratitude to my academic supervisors Dr. Sana Ben Hamida and Dr. Marc Baudry for supporting this international research experience as part of my journey in the Data Science for Social Sciences CMI master's degree.

Contents

Introduction	4
1 Host Organization	5
1.1 International Research Experience Program	5
1.2 The University of Oregon	7
1.3 The Rohlfslab	8
1.3.1 Presentation of the laboratory	8
1.3.2 Internship Description	9
2 Research Context and Problem Statement	10
3 Methodology and Resolution	13
3.1 Work environment	13
3.2 Data	14
3.3 Methods	15
3.3.1 Variable operationalization	16
3.3.2 Econometric Specifications	18
3.3.3 Estimation Code and Statistical Tooling	20
4 Results	21
4.1 Overview and Exploratory Analysis	21
4.1.1 Descriptive Statistics	21
4.1.2 Call Outcome Exploratory Analysis	22
4.2 Model 1: All Agencies Arrest Rate Impact	22
4.3 Model 2: Police Department Outcome	25
4.4 Model 3: Interaction Model by Call Nature	25
5 Conclusions and Perspectives	26
Appendix	28

I	Map of BTAA Universities	28
II	SPD call data table	29
III	CAD data table	31
IV	MCSLC data table	32
V	Final data table	33
VI	Parallel trends check	34
VII	Plots	35
VIII	International experience	42

Introduction

Thirty-seven years ago, the city of Eugene, Oregon was the birthplace of the very first Mobile Crisis Response Program (MCRP) in the USA. Pioneered by White Bird Clinic and the Eugene Police Department (EPD), CAHOOTS (Crisis Assistance Helping Out On the Streets) has been an alternative to police emergency response for non-violent behavioral health interventions, mental health crisis, and homeless street outreach. The CAHOOTS model, working hand-in-hand with the police department, aims at helping people in need and providing services outside of the police's or the emergency department's scope such as crisis de-escalation, transportation, non-emergent medical issues, drug and alcohol counseling, connection to shelters [1] etc. The program functions with a mobile two-person team consisting of a medic and a crisis worker trained in the mental health field. They are dispatched through the 911 emergency number alongside the police.

However, CAHOOTS was discontinued in Eugene on April 7 2025. As of today, the only MCRP running in the city, is the Mobile Crisis Service from Lane County (MCS-LC), launched on August 18, 2024. This service focuses on behavioral health emergencies as well. But unlike CAHOOTS, MCS-LC is not directly available through the 911 emergency line and is not automatically dispatched through the Computer-Aided Dispatch system. They also do not provide non-emergent medical aid as their team only includes mental health professionals [2].

In this context, the main purpose of this analysis is to investigate the changes in response (call outcome) to non-violent emergency calls, with a focus on arrests rates and the Mobile Crisis Services of Lane County after the discontinuation of CAHOOTS. Thus, we ask the question: how CAHOOTS' discontinuation impacted call outcome for non-violent crisis emergency response?

This report documents a summary of my research internship and the completion of this project within the Rohlfslab, at the University of Oregon, USA. The first part focuses on presenting the host organization and the general context into which this internship experience takes place. The second section is a description of the research question we focused on, then an overview of the state of the art in Data Science. The methodology section walks through the resolution of the problem statement. Finally, we will review the conclusions and perspectives of this study.

1 Host Organization

This 13-week internship is part of the International Research Experience Program (IREP)[4], jointly created through the Big Ten Academic Alliance (BTAA) and the CMI FIGURE network. This program allows CMI students from France to apply for an internship in one of the American universities member of the BTAA conference.

I got invited by Dr. Rori Rohlf to join the Rohlfslab at the University of Oregon, located in the Pacific Northwest, to work on Mobile Crisis Response Services in Eugene (United-States of America).

1.1 International Research Experience Program

The FIGURE network (Formation en InGénierie par des Universités de REcherche) [5] was created in France in 2013 as an association, originally initiated by Robert Chabbal, former director general of the CNRS. Its purpose is to develop a new model of engineering education within research universities, as an alternative to the French engineering school system, aligned with the international Master of Science in Engineering standard. The network currently includes around a hundred CMI programs across 23 universities in France and abroad, and holds the European EUR-ACE label (European Accredited). The partnership between the FIGURE network and the BTAA created the IREP program [4] as a way to facilitate short-term research experiences in the United States for CMI students, and give them an opportunity to immerse in a fully english-speaking academic environment. This is a competitive program where applicants are selected based on their academic record, their research interests, and the availability of matching opportunities in partner labs.

The Big Ten Conference [6] is a collegiate athletic conference founded in 1896, consisting of ten prominent universities, which later expanded to 18 member institutions and two affiliate institutions. The group is affiliated with Division I of the National Collegiate Athletic Association, and is based in Rosemont, Illinois. The conference includes the following universities (map overview available in Figure 1):

- University of Illinois Urbana-Champaign
- Indiana University

- University of Iowa
- University of Maryland
- University of Michigan
- Michigan State University
- University of Minnesota
- University of Nebraska-Lincoln
- Northwestern University
- The Ohio State University
- University of Oregon
- Pennsylvania State University
- Purdue University
- Rutgers University
- University of California, Los Angeles (UCLA)
- University of Southern California (USC)
- University of Washington
- University of Wisconsin-Madison

The academic counterpart was created in 1958 under the name "Committee on Institutional Cooperation" which became the Big Ten Academic Alliance [7] in 2016 and has a network of over 700,000 students.

Coordinated by Laurel Rosch and based at the University of Illinois Urbana-Champaign, the IREP experience allows CMI students to conduct research at a leading Big Ten university in the United States. The program helps them explore research interests in STEM fields and advance their academic or private sector careers. It is a short-term, unpaid research position that does not award academic credit. Students enrolled in the program can enter US territory as visiting scholars with a J-1 visa that cannot extend beyond six months.

Applications from the students, that opened in April 2025, include a Curriculum Vitae, a cover letter, and a statement of research interest. This paperwork allows the program coordinators from partner universities to match applicants with potential labs and researchers in their field of interest.

Notifications to applicants were sent in October 2025 for a placement in Spring/-Summer 2026. I received two placement offers, one from the SyNSec lab (Systems and Network Security research group) at Pennsylvania State University, and one from the Rohlfslab at the University of Oregon. I chose the latter because the research topic was more aligned with my professional goals and personal interests.

1.2 The University of Oregon

The University of Oregon (UO) [9][10], founded in 1876, is a tier 1 comprehensive public research university in Eugene, Oregon, United States. The 119 hectares main campus is located along the Willamette River and provides a broad range of academic opportunities, with more than 420 undergraduate and graduate degree programs across diverse fields. It is organized into several schools and colleges, including the College of Arts and Sciences, Lundquist College of Business, School of Law, College of Design, and College of Education. These units support both foundational and specialized training.

The research environment at UO is characterized by a strong commitment to innovation and societal impact. The university is a host to multiple research centers and institutes that encourage interdisciplinary collaboration, including initiatives focused on sustainability, data science, neuroscience, and global development. Faculty and students are actively engaged in research supported by federal agencies, private foundations, and industry partnerships.

In addition to its academic and research strengths, the University of Oregon maintains a commitment to community engagement and experiential learning. Through partnerships with local, regional, and global organizations, the university provides students with opportunities to apply their knowledge in real-world contexts.

1.3 The Rohlfslab

1.3.1 Presentation of the laboratory

The Rohlfslab is a computational biology research group that started at San Francisco State University, then relocated at the University of Oregon in 2022. It is part of the Institute for Ecology and Evolution (IE²) as well as the School of Computer and Data Sciences. Led by Dr. Rori Rohlf, the laboratory focuses on population genetics, evolutionary biology, and the development of quantitative methods.

The IE² is a research institute for "ecology, evolution, genetics, and genomics, with a focus on understanding how biological diversity emerges, adapts, and responds to environmental change" [11]

The UO School of Computer and Data Science (SCDS) is an academic unit that combines training in computational theory, data analysis, and statistical engineering. The school hosts undergraduate and graduate programs focused on quantitative methods such as math, statistics, and machine learning, as well as ten different domain areas to apply those skills to [12]. Research within SCDS covers foundational areas like network architecture and database optimization, as well as applied domains such as bioinformatics and geographic data analytics. SCDS operates as an independent division within the university infrastructure, partnering with various research centers to support computational infrastructure and baseline technical literacy across disciplines.

The main area of research involves modeling gene expression evolution. The laboratory designs statistical frameworks to distinguish between natural selection and neutral genetic drift. This work helps clarify the functional mechanisms behind evolutionary changes over time.

Although the laboratory is focused on biology and genetics, another area of research is oriented towards social justice and data science with projects like the CAHOOTS impact evaluation.

The laboratory group currently consists of postdoc researcher Dr. Carolina Adam, postbac researchers Myco Torres and Nathan Burton, and undergraduate researchers Alicia Bierly, Melody Chang, Gisselle Quiroz and Sebastián Iturbe. Active projects include the tracking of tandem repeat (repetitive DNA sequences) evolution across primate genomes [13], led by Dr. Carolina L. Adam. She uses comparative and

population genomics approaches to analyze how repetitive DNA contributes to evolutionary change over long timescales. This analysis connects directly with narrower student projects, such as the investigation of the role of tandem repeats in adaptive archaic introgression across the human genome, led by undergraduate researches Alicia Bierly and Melody Chang [14], tandem repeat mutation simulation [15], led by undergraduate researchers Luis Fernandez-Luna, Sebastián Iturbe and Gisselle Quiroz, and finally the CAHOOTS project, led by post-bac researcher Nathan Burton.

1.3.2 Internship Description

The internship took place on site, over the course of three months, from April to June 2026. The experience mainly consisted of working on the CAHOOTS quantification research project. I attended the Data Science in Action class twice a week, taught by Dr. Rori Rohlf. It is a research based course where students have to conduct a data science project from start to finish, using the CAHOOTS data. I also had a weekly meeting with collaborator Nathan Burton and Rori Rohlf where we discussed findings and the progress made towards the research project.

Regarding life at the laboratory, another weekly meeting was held with other members. The topic of each meeting changed each time. Some activities we did included poster and research presentation, research paper reading and talk, and presentation practice for future conferences. I also attended weekly biology seminars hosted by the IE², the Undergraduate Research Symposium to support Alicia and Melody's poster presentation, and other one-off research events hosted by the University. For instance, I presented at the Global Studies Institute's research symposium where I had the opportunity to talk about my work here and learn about other researchers' work they did abroad.

2 Research Context and Problem Statement

Emergency response systems in the United States face heavy pressure to address crises related to mental health, homelessness, addiction and other related issues. The 911 emergency line traditionally sends police officers to handle these calls, even if no crime is committed [16]. A response system like this can cause problems, such as higher costs, unnecessary crisis escalation and arrests. Many cities have put in place MCRPs to solve this issue [17].

After the murder of George Floyd in May 2020, the police reform and response to mental health related crisis has been brought back into the public eye, with a nationwide debate on law enforcement budgets and redirection toward social and mental health services. The oldest MRCP in the United States is the CAHOOTS program, created in 1989 in Eugene, Oregon. It began as a partnership between the Eugene Police Department (EPD) and the White Bird Clinic, and has served as a national model for Mobile Crisis Response Programs (MCRPs) for over three decades. By deploying mental health professionals and social workers instead of police for non-violent emergency calls, MCRPs aim to provide a civilian and empathetic alternative that addresses behavioral health crisis without the need for traditional law enforcement intervention. CAHOOTS was frequently cited in that debate as a proof-of-concept for civilian crisis response [18]. Senator Ron Wyden and Representative Peter DeFazio introduced the CAHOOTS Act at the federal level, explicitly modeled on the Eugene program. This proposition includes allowing "state Medicaid programs certain community-based mobile crisis intervention services for individuals experiencing a mental health or substance-use disorder crisis outside of a facility setting" [19]. Congress also passed in 2020 the National Suicide Hotline Designation Act, creating the 988 line, designed for suicide prevention and mental health crisis response in the U.S. [20]. It was launched in July 2022 and is available in every community, and is a sign of federal acknowledgment that crisis should not default to law enforcement.

Going back to CAHOOTS, it is a two-person team structure consisting of one medically trained person, and a social worker. They don't have any law enforcement power and are unarmed [1]. They were dispatched through the 911 CAD system alongside EPD, so call-takers could choose to send CAHOOTS instead of or with the police depending on the nature of the call. Integration was a key structural feature, distinguishing them from newer MCRPs like MCS-LC. Their scope of service includes crisis de-escalation, welfare checks, transportation to shelters or treatment, non-emergent

medical aid, substance use counseling, outreach to homeless individuals. They are explicitly not designed for violent calls or situations requiring law enforcement, and they reserve the right to call the police as needed [1], even though the goal of the program is to give an appropriate response to this kind of emergencies, and to lower law enforcement costs while helping relieving the police’s workload so they can focus on more important interventions.

The question of how effective these programs actually are at diverting calls away from police has been a subject of considerable debate. The EPD Crime Analysis Unit published an internal evaluation of the CAHOOTS program using Eugene’s CAD data, estimating a diversion rate between 3% and 8% [21]. These figures became widely cited in policy debates around MCRP replication, but the methodology underlying them was never formally validated. This administrative downward bias stemmed from critical methodological errors, including the exclusion of core service categories (call natures) like public assist and client transportation. Recent data science literature resolves this controversy by moving past static descriptive statistics and introducing quasi-experimental designs. Nathan Burton and Rori Rohlf, from the Rohlfslab at the University of Oregon, directly addresses this by re investigating this research question using the same CAD dispatch logs (the same data used in this analysis), reconstructing their assumptions, and testing how sensitive the diversion rate estimate is to different methodological choices. By analyzing historical service hour expansions as a natural experiment, researchers isolated three distinct metrics. The substitution diversion rate measures direct police replacement and stands at 18%. The prevention adjusted diversion rate accounts for early de-escalation that stops future calls from occurring and reaches 23%. Finally, the overlapping mandate diversion rate shows that the civilian team handles half of all mutually addressable calls within the shared scope of both agencies [16].

Rigorous econometric evaluations confirm that integrating these civilian teams produces major institutional shifts. Difference in differences designs tracking service expansions show that a civilian crisis response causally reduces the probability of a call resulting in an arrest by 24 percentage points. This reduction is driven entirely by a drop in non-criminal mental health holds and detoxification detentions. Law enforcement officers often utilize these holds due to a lack of specialized psychiatric diagnostic training. Plus, the presence of civilian teams has no impact on outstanding warrant arrests, confirming that dispatchers target the teams accurately to low-risk scenarios. At the same time, alternative crisis response improves overall access to healthcare.

According to research from Jonathan Davis, from the University of Oregon, the presence of a mobile crisis team increases the likelihood of emergency medical services arriving on scene by 26 percentage points and elevates emergency room transports by 8.8 percentage points. These metrics prove that civilian response resolves incidents through medical care rather than criminal justice channels [17].

The emergency response landscape in Eugene experienced a profound structural disruption through two consecutive policy shifts. On August 18, 2024, Lane County launched the Mobile Crisis Services of Lane County to handle behavioral emergencies. This county program matches a different operational model. It relies exclusively on mental health staff without medical personnel, and it operates outside the 911 computer aided dispatch network. Next, on April 7, 2025, the city completely discontinued the CAHOOTS within Eugene due to municipal budget issues. The adjacent city of Springfield chose to maintain its civilian service hours, creating a clear geographical control group. This sudden loss of an integrated civilian model provides a critical data science opportunity. This report utilizes a difference in differences framework to evaluate how the discontinuation impacted call outcomes and arrest rates for non-violent emergencies across Eugene.

This research project, as a complement to the research currently led by Nathan Burton at the Rohlfslab, will focus on investigating the changes in Eugene's response to emergencies related to homelessness, addiction and mental health after CAHOOTS' discontinuation, with a focus on arrest rates and the role of the EPD and MCS-LC after this event. The research question we will be focusing on is: *how did the discontinuation of CAHOOTS and the concurrent operation of MCS-LC affected call outcomes for non-violent crisis emergencies in Eugene?* New research in progress from Nathan Burton also includes investigating the changes in call priority assigned by dispatchers for CAHOOTS call types, and continuation of the gap analysis to see how non-violent emergency crisis are now handled in Eugene.

3 Methodology and Resolution

To rigorously evaluate the structural impact of the discontinuation of CAHOOTS in Eugene on April 7, 2025, and the simultaneous reliance on Mobile Crisis Services Lane County (MCS LC), this study employs a multi-period Difference-in-Differences (DiD) framework. This section details the data harmonization pipeline, the spatial-temporal boundaries of the analysis, and the formal econometric specification used to test the parallel trends assumption.

3.1 Work environment

This project was conducted using R (version 4.5.1). This programming language, additionally to being the one I am most comfortable working with, is a natural choice when it comes to this kind of social science quantitative analysis. It was designed for statistical computing and data analysis, unlike Python which is more general-purpose. Because we use the Differences-in-Differences, and other econometrics methods, R is particularly well suited and documented for this, with packages like `fixest` and `did` that are actively maintained by the community. It also offers strong visualization options with `ggplot2`, which is standard in academia. The R Markdown (`.Rmd`) format is also very convenient for allowing us to combine code, output, and text in a single document, making the analysis reproducible and self-documenting.

This project is fully available on GitHub [22] for version control. It includes the whole data processing pipeline, in `.R` scripts, and the code for the analysis in a `.Rmd` file used simultaneously for the analysis and documentation. For reproducibility, library dependencies are managed using `renv`. More information on how to clone and use the project are available in the `README` documentation that you can find on the repository.

Some key libraries used to carry out the analysis include `tidyverse`, and some of its dependent packages such as `dplyr` for data manipulation, and `ggplot2` for visualization. It is a collection of open source packages designed for data science, introduced by Hadley Wickham, based on his tidy data principles [23]. I also used causal inference and econometrics libraries like `fixest` and `did`, along with `stargazer` for display and regression formatting.

3.2 Data

For this project, three administrative data sources were used. Location information was removed from the data to preserve anonymity. We have Law Enforcement and CAHOOTS call data for the city of Eugene, coming from the 911 Computer-Aided Dispatcher (CAD), as well as for the adjacent city of Springfield. Additionally, we use the call data logs for the MCSLC program, with a focus on Eugene’s calls. To facilitate the analysis, we cleaned and transformed the data to merge it all in one data frame, keeping only the relevant information.

SPD call data: The Springfield Police Department (SPD) data consists of three datasets that were merged together:

- Call for Services: call dispatch logs for CAHOOTS and the SPD from 2015 to 2025.
- Responding Units: units dispatched for every call logged in the first dataset.
- Call outcome: close codes for every call logged in the first dataset.

The consolidated dataset, with 14 columns and 567,655 rows, tracks variables such as the `timestamp`, a unique identifier for each call, as well as information such as the `city`, `nature`, `close_code` of incident, its `priority`, and temporal variables like `dispatch_time`, `arrival_time`.

To make it usable for further analysis, we processed it to identify calls where CAHOOTS was dispatched using regular expressions and unit code identification, and created binary columns, `SPD` and `CAHOOTS`, with value equals 1 if the agency was involved, 0 otherwise. The full descriptive table for the processed data can be found at Appendix II.

CAD call data: For the city of Eugene, we use the 911 emergency line’s Computer-Aided Dispatch call data. Similarly to the SPD call data, this dataset includes a `timestamp`, a unique identifier, and additional information on the incident.

With 17 columns and 1,446,014 rows, it is composed of the call logs and dispatches for the Eugene Police Department (EPD) and for CAHOOTS in Eugene. We identi-

fied CAHOOTS calls using regular expressions for 'CAHOOTS' string in call nature and outcome, and the unit number used by dispatchers[24]. We then created two binary columns, **EPD** and **CAHOOTS**, to indicate their involvement for each call. See Appendix III for the descriptive table.

MCSLC call data: We are also using the Mobile Crisis Services of Lane County's call logs data. It tracks crisis intervention, focusing on response timelines, and clinical outcome for mobile crisis units. This dataset also includes a **timestamp**, a unique identifier, and additional information on the incident. Additionally to the dispatch, arrival, and clear time, durations are included in this dataset.

Consolidated dataset: For a more efficient use of the data throughout the analysis, we merged all 3 data sources into a unique dataset. To make sure the identifier stays unique, we added the source into each one (e.g. "cad_XXXXXX"). The final dataset has a binary column for every agency possible (SPD, EPD, CAHOOTS, and MCSLC), set to 1 if involved in the call, 0 if not. Only relevant information regarding the call outcome analysis was kept such as **nature** and **outcome** is in the dataset. See Appendix V for a detailed overview of the table. It has 1,601,142 rows and 12 columns. It is important to note that we kept only dispatched calls. Logged calls with no unit sent to the scene are excluded from the analysis.

3.3 Methods

Our main focus is on call outcome, and especially, but not limited to, arrests. In this part, I will explain how I calculated new variables (outcome rates), the methods for the exploratory analysis and how I constructed the Differences-in-Differences models used to quantify the changes in Eugene's emergency response after CAHOOTS stopped services.

Three models were used in this project. I first looked at arrest rates for all agencies in Eugene and Springfield before and after CAHOOTS stopped, then I focused on EPD calls only to see how outcomes changed for this particular agency. Next, I explored call types by breaking down outcomes for each to see which nature was more impacted by the service discontinuation.

3.3.1 Variable operationalization

Using the consolidated final dataset, I calculated the arrest rate using regular expressions to detect *arrest* and *juvenile taken into custody* outcomes, as it does not appear the same way in every data source. The same process was applied to other common outcomes for CAHOOTS call types I looked into, such as **Welfare Checks**, or **Emergency Department** interventions. For the first model, I only filtered call natures to match the most frequent CAHOOTS calls. I added a filter on the agency to keep only EPD and SPD (control group) calls for the second model, focused on police outcomes.

To isolate the true systemic shift in how non-violent crises are processed, the raw, individual-level emergency logs are filtered and aggregated using a controlled matching framework across three dimensions:

Target Call Scopes I chose to focus on CAHOOTS' most frequent observed call natures found across the CAD and SPD datasets, as to not bias the data by including call types trends that CAHOOTS never handled before. The fourteen following natures are the ones used for analysis:

- Public Assist
- Check Welfare
- Transport
- Suicidal Subject
- Intoxicated Subject
- Disorderly Subject
- Found Syringe
- Criminal Trespass
- Disoriented Subject
- Dispute
- Subject Down

- Disorderly Juveniles
- Suspicious Conditions
- Suspicious Subject

Temporal Aggregation The outcome variables are aggregated into a discrete weekly aggregation panel dataset spanning the treatment group (Eugene) and the control group (Springfield), using a calendar-week floor transformation:

$$\text{Week}_t = \lfloor \text{Timestamp}_i \rfloor_{\text{week}} \quad (1)$$

Weekly aggregation smooths out individual day-of-the-week volatility while allowing sufficient reliable trend estimation.

Outcome Variables The main dependent variable is the weekly arrest rate ArrestRate_{ct} , defined as the share of eligible dispatched calls in city c during week t that resulted in an arrest:

$$\text{ArrestRate}_{ct} = \frac{\sum_{i=1}^{n_{ct}} \text{Arrest}_{ict}}{n_{ct}} \quad (2)$$

where n_{ct} is the total number of eligible calls in city c during week t . The same formula is applied to other outcomes in the third model, simply replacing the arrest indicator with the outcome of interest.

Since the arrest rate is a proportion computed over a denominator that varies week to week, lower-volume weeks produce noisier estimates. To account for this, all regressions are weighted by $w_{ct} = n_{ct}$, giving more influence to weeks with higher call volumes. The key variables used in the models are summarized in Table 1.

Table 1: Variables used in the panel models

Variable	Construction in R	Role in the model
Unit of observation	<code>city × week</code>	Panel structure
Treated_c	<code>as.integer(city == "Eugene")</code>	Treatment group indicator
Post_t	<code>as.integer(timestamp >= "2025-04-07")</code>	Post-discontinuation indicator
ArrestRate_{ct}	<code>sum(arrest) / n()</code>	Dependent variable
Weight	<code>weights = n_calls</code>	Correction for varying call volume

3.3.2 Econometric Specifications

Parallel Trends Before running the DiD models, I verified the parallel trends assumption, which requires that arrest rates in Eugene and Springfield were evolving similarly prior to CAHOOTS' discontinuation. This was tested using data from January 2023 through April 7, 2025, with the following interaction model estimated on the pre-treatment period only:

$$\text{ArrestRate}_{ct} = \alpha + \beta_1 t + \beta_2 \mathbb{I}[\text{City} = \text{Eugene}]_c + \beta_3 (t \times \mathbb{I}[\text{City} = \text{Eugene}]_c) + \varepsilon_{ct}, \quad \forall t < t^* \quad (3)$$

$$\begin{cases} H_0 : \beta_3 = 0 & \text{(parallel trends assumption holds)} \\ H_1 : \beta_3 \neq 0 & \text{(trends diverge dynamically)} \end{cases} \quad (4)$$

Where:

- $\mathbb{I}[\text{City} = \text{Eugene}]_c$ is a dummy variable equal to 1 for Eugene (treatment) and 0 for Springfield (control).
- t is a weekly time index, set to 0 at the week of the discontinuation, negative before, and positive after.
- β_3 is the interaction coefficient. A non-significant estimate (fail to reject H_0) indicates that the two cities' arrest rates were trending in parallel before the treatment, which validates the DiD design.

The same test was applied for each outcome variable used in the other two models. If the assumption did not hold for a given outcome, the DiD result is reported descriptively rather than as a causal estimate.

Difference-in-Differences (DiD) Specifications Once the parallel trends assumption was verified, three DiD models were estimated.

Model 1: All Agencies Arrest Rate Impact The first model estimates the overall effect of CAHOOTS' discontinuation on arrest rates, pooling calls from all

agencies:

$$\text{ArrestRate}_{ct} = \alpha + \gamma_1 \mathbb{I}[\text{City} = \text{Eugene}]_c + \gamma_2 \text{Post}_t + \delta_{\text{DiD}} (\mathbb{I}[\text{City} = \text{Eugene}]_c \times \text{Post}_t) + \eta_{ct} \quad (5)$$

Where $\text{Post}_t = \mathbb{I}[t \geq 0]$ flags weeks after April 7, 2025, and δ_{DiD} is the coefficient of interest, capturing the change in Eugene's arrest rate relative to Springfield after the discontinuation.

Model 2: Police Outcomes Only The second model asks whether police-handled calls specifically saw a shift in outcomes after CAHOOTS stopped. The sample is restricted to EPD (Eugene) and SPD (Springfield) calls:

$$\text{ArrestRate}_{ct}^{\text{Police}} = \alpha_{\text{PD}} + \theta_1 \mathbb{I}[\text{City} = \text{Eugene}]_c + \theta_2 \text{Post}_t + \delta_{\text{DiD}}^{\text{PD}} (\mathbb{I}[\text{City} = \text{Eugene}]_c \times \text{Post}_t) + \nu_{ct} \quad (6)$$

The denominator n_{ct}^{Police} counts only police-handled calls, and $\delta_{\text{DiD}}^{\text{PD}}$ measures whether EPD calls were more likely to result in arrests after CAHOOTS was no longer available to handle these types of incidents.

Model 3: Breakdown by Call Nature The third model tests whether the effect differs across call types. Using the same DiD structure as Model 1 (and extended to other outcomes beyond arrests), I add interaction terms for each call nature, with *Check Welfare* as the reference category:

$$\begin{aligned} Y_{c,n,t} = & \alpha + \beta_1 \mathbb{I}[\text{City} = \text{Eugene}]_c + \beta_2 \text{Post}_t + \gamma_n \text{Nature}_n \\ & + \beta_3 (\mathbb{I}[\text{City} = \text{Eugene}]_c \times \text{Post}_t) \\ & + \beta_4 (\mathbb{I}[\text{City} = \text{Eugene}]_c \times \text{Nature}_n) + \beta_5 (\text{Post}_t \times \text{Nature}_n) \\ & + \delta_n (\mathbb{I}[\text{City} = \text{Eugene}]_c \times \text{Post}_t \times \text{Nature}_n) + \varepsilon_{c,n,t} \end{aligned} \quad (7)$$

Where:

- $Y_{c,n,t}$ is the weekly outcome rate for city c , call nature n , and week t .
- Nature_n is a set of dummy variables for each call type, excluding the reference category.
- β_3 gives the DiD estimate for the reference nature (*Check Welfare*).
- δ_n measures how the effect for nature n deviates from the reference.
- The total DiD effect for nature n is $\beta_3 + \delta_n$.

An F -test under the joint null $H_0 : \delta_1 = \delta_2 = \dots = \delta_k = 0$ checks whether the treatment effect is uniform across call types or whether certain natures were more affected than others.

3.3.3 Estimation Code and Statistical Tooling

All three models are estimated in R using the `lm` function, with `fixest` as an alternative for faster computation via its `feols` engine. All regressions are weighted by weekly call volume as described above. Standard errors are clustered at the city-week level to account for serial correlation over time. Results are formatted and displayed using the `summary` and `stargazer` packages.

4 Results

4.1 Overview and Exploratory Analysis

4.1.1 Descriptive Statistics

Data Summary and Global Call Volume The merged dataset comprises 1,601,142 dispatched calls across four agencies (EPD, SPD, CAHOOTS, and MCSLC) from 2015 to 2025. Eugene accounts for the vast majority of observations ($n = 1,031,525$ calls), followed by Springfield ($n = 568,239$ calls).

During the analysis period, Eugene Police Department (EPD) responded to 905,197 calls, while CAHOOTS handled 125,652 calls over its operational period. Springfield Police Department (SPD) responded to 527,491 calls, with CAHOOTS handling 52,508 calls in that jurisdiction. The newly launched Mobile Crisis Services Lane County (MCS LC) recorded 7,097 dispatches in Eugene and 710 in Springfield, reflecting its recent implementation as of August 18, 2024.

Most Frequent Call Natures CAHOOTS interventions were concentrated among a small set of call types. The two most prevalent call natures were *Public Assist* ($n = 57,748$ calls, 46% of all CAHOOTS calls) and *Check Welfare* ($n = 57,605$ calls, 46%), accounting for approximately 92% of CAHOOTS dispatches. Other frequently handled call natures included *Transport* ($n = 20,922$, 17%), *Suicidal Subject* ($n = 11,051$, 9%), and *Intoxicated Subject* ($n = 3,512$, 3%) (see figure 3). This distribution reflects CAHOOTS' specialization in welfare checks, community engagement, and non-violent behavioral health interventions.

In contrast, MCS LC's call portfolio during the overlap period (August 18, 2024–April 7, 2025) centered on behavioral and mental health crises. The most common presentations were *Agitation or Disruptive Behavior*, *Disorganized Behavior*, and *Difficulty Functioning*. This difference in call nature distributions reflects the distinct dispatch pathways and clinical mandates of the two services (see figure 4).

Temporal Evolution and Service Transitions Between 2015 and 2025, CAHOOTS maintained a relatively stable operational volume. MCS LC's launch on Au-

gust 18, 2024, marked the beginning of a transition in Eugene’s crisis response infrastructure. During the eight-month overlap period (August 18, 2024 — April 7, 2025), CAHOOTS completed 9,038 dispatches while MCS LC recorded 1,116 dispatches. Following CAHOOTS’ discontinuation on April 7, 2025, MCS LC’s call volume increased substantially to 5,894 dispatches over the subsequent 254-day period (See figure 2).

4.1.2 Call Outcome Exploratory Analysis

CAHOOTS interventions predominantly resulted in community-based outcomes: 64.4% of calls were closed as *Community Support* ($n = 5,817$ calls), with the outcome *Assisted* representing the most frequent disposition. An additional 27.6% of calls ($n = 2,491$) resulted in *No Contact* outcomes, primarily comprising *Unable to Locate* ($n = 2,063$) and *Gone on Arrival* ($n = 824$) (see figure 5). Arrests were exceptionally rare during CAHOOTS operations, occurring in only 0.02% of calls ($n = 2$).

In comparison, MCS LC exhibited a similar but distinct outcome profile. Among the 1,116 dispatches during the overlap period, 76.8% ($n = 857$) resulted in *Remained in Community*. Emergency Department referrals occurred in 19.4% of cases ($n = 216$), reflecting higher acuity presentations or treatment engagement barriers. Arrests, while still rare, were slightly more frequent than CAHOOTS at 0.6% ($n = 7$), likely reflecting the presence of police involvement in certain calls.

Notably, both services achieved their highest community support rates for low-risk call types such as welfare checks and public assistance, while emergency department referrals and police involvement increased for higher-acuity presentations including suicidal ideation and harm-risk scenarios.

4.2 Model 1: All Agencies Arrest Rate Impact

To rigorously assess the impact of CAHOOTS’ discontinuation on police response outcomes in Eugene, we employed a difference-in-differences (DiD) methodology, with Springfield serving as a control group. This approach isolates the causal effect of the service discontinuation by accounting for pre-existing trends and baseline differences between the two cities.

Parallel Trends Verification A prerequisite for valid DiD inference is the parallel trends assumption: arrest rates in Eugene and Springfield must have evolved similarly prior to CAHOOTS’ discontinuation. Before running the statistical test, I plotted arrest rates by city with a 4 week rolling average, suggesting the presence of parallel trends (see figure 6). We then tested this assumption using data from 2023 through April 7, 2025 (the discontinuation date).

Table 2: Parallel Trends Statistical Testing

	<i>Dependent variable:</i>
	arrest_rate
week_num	0.0001*** (0.00003)
treated	0.012*** (0.002)
week_num:treated	−0.00003 (0.00004)
Constant	0.019*** (0.002)
Observations	236
R ²	0.271
Adjusted R ²	0.262
Residual Std. Error	0.186 (df = 232)
F Statistic	28.746*** (df = 3; 232)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

The primary parameter of interest for validation is the coefficient on the interaction term **week_num** × **treated**, which captures the differential temporal trend between the treatment group (Eugene) and the control group (Springfield) prior to the policy shock. This estimate is small and statistically non-significant ($\hat{\beta}_3 = -0.00003$, $p > 0.1$), indicating that the baseline arrest rates for both cities evolved along parallel trajectories with no systemic temporal drift prior to the discontinuation of CAHOOTS.

In contrast, the indicator for the treatment group (**treated**) is positive and highly significant ($\hat{\beta}_2 = 0.012$, $p < 0.01$), revealing a stable baseline level difference where the arrest rate in Eugene is approximately 1.2 percentage points higher than in Spring-

field across identical call natures. The common time variable (`week_num`) exhibits a slight but statistically significant upward trend common to both baseline environments ($\hat{\beta}_1 = 0.0001$, $p < 0.01$). Because the pre-treatment slopes do not statistically diverge, the parallel trends assumption holds, justifying the use of Springfield as a valid counterfactual.

Difference-in-Differences Estimates Results revealed a statistically significant and substantial increase in arrest rates following CAHOOTS' discontinuation:

Table 3: DiD, Effect of CAHOOTS Discontinuation on Arrest Rate (Weekly)

	<i>Dependent variable:</i>
	Arrest Rate
Eugene (Treated)	0.010*** (0.001)
Post-Discontinuation	-0.002 (0.003)
DiD Estimator (Treated $\times$ Post)	0.036*** (0.003)
Constant	0.025*** (0.001)
Observations	314
R ²	0.653
Adjusted R ²	0.649
Residual Std. Error	0.207 (df = 310)
F Statistic	194.265*** (df = 3; 310)

Note:

*p<0.1; **p<0.05; ***p<0.01
OLS weighted by weekly call volume.

This coefficient indicates that arrest rates for calls of types typically handled by CAHOOTS increased by 3.6 percentage points in Eugene relative to Springfield following CAHOOTS' discontinuation.

4.3 Model 2: Police Department Outcome

For a more detailed analysis to see how police call outcome changed after CAHOOTS stopped servicing Eugene, I tried answering the question: did the rate of specific outcome change in Eugene relative to Springfield for law enforcement after April 7, 2025?

I used the same method as model 1. Parallel trends assumption did not pass for every outcome, which means that we cannot use the DiD model for all. Outcomes for which the assumption held were still put into a DiD model, but results were mostly non-significant. Instead, I decided to limit the results to a descriptive analysis in figure 7, which displays variation in arrest rates by call nature after CAHOOTS' discontinuation for the EPD. While every outcome saw its arrest rate increase, we cannot say it is due to CAHOOTS stopping service in Eugene.

4.4 Model 3: Interaction Model by Call Nature

For the last model, I ran the same parallel trend testing and DiD model, adjusting the data by filtering it call nature by call nature to break down the results and see if outcomes were impacted differently. Most DiD results did not pass but we still see a significant percentage point increase for **Criminal Trespass** (+5.6 points), and **Disorderly Subject** (+4.05 points). Full visual results are available in figure 8.

The interaction model corroborates our findings from the first DiD model. Coefficients are interpreted relatively to the **Check Welfare** outcome. The full result table is available at figure 9.

5 Conclusions and Perspectives

This report examined how Eugene’s response to non-violent crisis calls changed after the discontinuation of CAHOOTS on April 7, 2025. Using administrative dispatch data from Eugene and Springfield, I first described the structure of the calls handled by CAHOOTS and MCS LC, then estimated the short-run effect of CAHOOTS’ closure on arrest outcomes through a Difference-in-Differences model. The results point to a shift in how these calls were resolved after the service stopped, but I recommend staying careful interpreting these results, as we do not find many significant results, and data is subjected to dispatcher and researcher biases.

The main finding is that arrest rates increased significantly in Eugene relative to Springfield after the discontinuation of CAHOOTS. The parallel trends assumption was not rejected in the pre-treatment period, which supports the use of Springfield as a comparison city. Under this identification strategy, the DiD estimates show an increase of 3.6 percentage points in the arrest rate for the set of call natures typically handled by CAHOOTS.

The exploratory analysis also helps interpret this change. CAHOOTS primarily handled calls related to public assistance, welfare checks, transport, and suicidal subject situations, and most of these interventions ended with community-based outcomes. MCS LC, although smaller and more recent, showed a different distribution of call natures and outcomes, with a larger share of calls remaining in the community but also a non-negligible share being referred to the emergency department.

More broadly, the findings suggest that specialized mobile crisis response plays an important role in limiting arrests in non-violent emergency situations. While this study does not claim to measure all consequences of CAHOOTS’ discontinuation, it provides evidence that the change had immediate implications for public safety response in Eugene.

Future work could extend this analysis in several directions. It would be useful to examine additional outcomes such as emergency department referrals, transport outcomes, and no-contact cases. Additionally a longer post-discontinuation period would make it possible to better assess whether the observed effects persist or stabilize over time. Finally, combining research currently being conducted by Nathan Burton investigating the gap left by CAHOOTS, and how it is filled by agencies still

in service would provide a more complete picture of how mobile crisis systems affect both immediate outcomes and broader patterns of emergency response.

Appendix

I Map of BTAA Universities

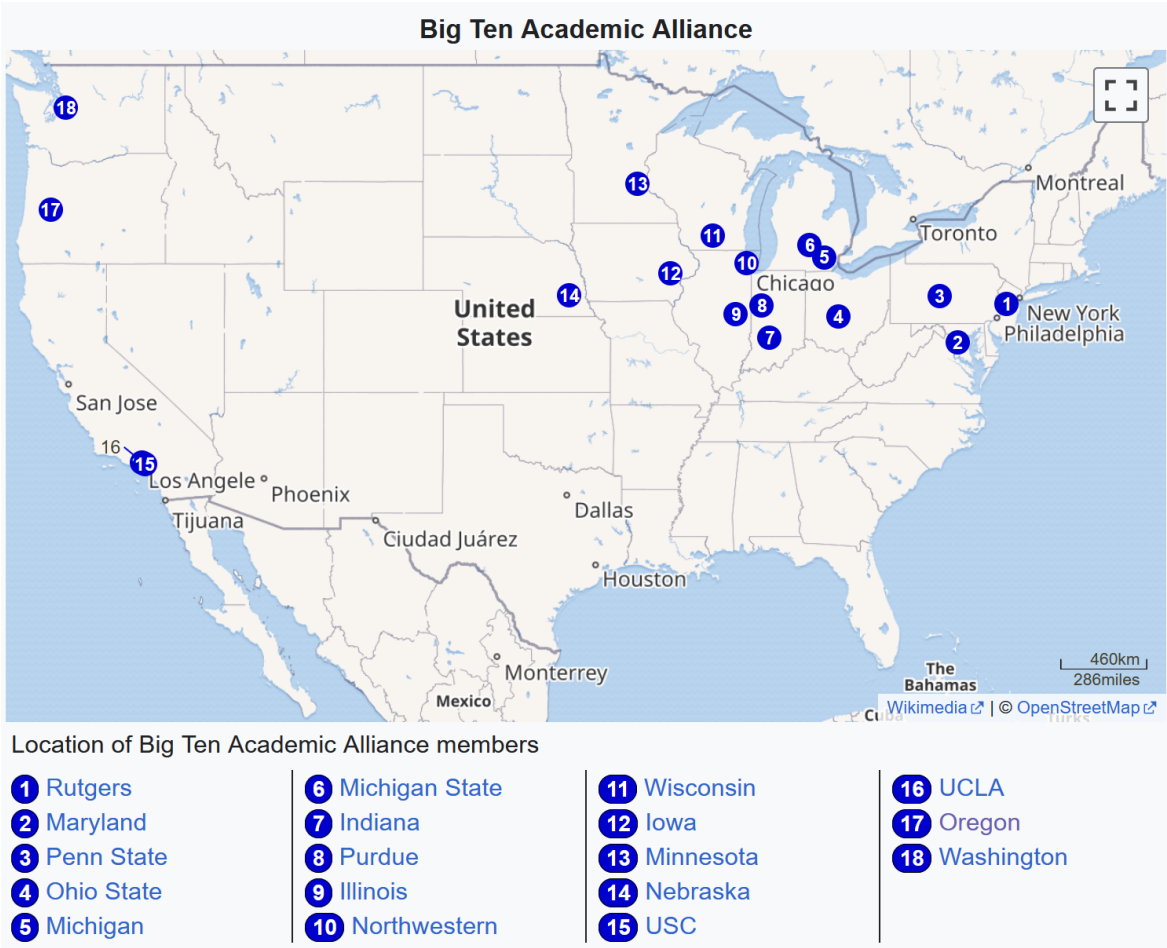


Figure 1: Map of BTAA Universities participating in the IREP program

II SPD call data table

Variable	Type	Example Value	Description
timestamp	datetime	2015-01-01 01:16:21	Date and time.
incident_id	double	15000074	Unique identifier.
agency	factor	SPD	Agency in charge.
city	factor	Springfield	City.
nature_code	factor	ALMAUD	Call type code.
nature_final	factor	AUDIBLE ALARM	Call type final classification.
close_code	factor	BLDS	Call outcome code.
close_code_definition	factor	Building Check Secure	Call outcome description.
priority	factor	3	Numerical priority level assigned.
prime_unit	character	1S18	Primary unit assigned to respond to the incident.
units	character	1S18, 1S21, K94	List of all units dispatched to the scene.
dispatch_time	time	01:18:00	Time at which units were dispatched.
arrival_time	time	01:21:00	Time at which the first unit arrived at the scene.
clear_time	time	01:32:00	Time at which the incident was marked as completed.
call_source	factor	PHONE	Origin or method of the incoming call.

III CAD data table

Variable	Type	Example Value	Description
timestamp	datetime	2015-01-01 00:00:00	Date and time.
incident_id	double	15000001	Unique identifier.
agency	factor	EPD	Agency in charge.
service	factor	LAW	Type of service (e.g., Law Enforcement, Fire, Other).
call_source	factor	SELF	Origin of the call.
nature	factor	PERSON STOP	Nature (call type) or reported motive of the incident.
close_code	factor	ASST	Call type code
closed_as	factor	ASSISTED	Call outcome description.
priority	factor	6	Priority level assigned to the incident ranging from 1 (highest) to 9 (lowest).
dispatch_time	time	00:00:00	Time until dispatch for the incident location.
arrival_time	time	00:00:00	Time until first unit's arrival on the scene.
clear_time	time	00:03:37	Time when the incident was marked as cleared or completed.
dispatch	factor	1	Binary indicator specifying whether a unit was dispatched.
arrival	factor	1	Binary indicator specifying whether a unit arrived on the scene.
prime_unit	factor	_5E48	Identifier of the primary unit assigned to the incident.
nb_units_dispatched	double	1	Total number of units dispatched.
nb_units_arrived	double	1 31	Total number of units arrived on scene.

IV MCSLC data table

Variable	Type	Example Value	Description
timestamp	datetime	2025-12-17 02:57:00	Date and time of log.
incident_id	double	13209	Unique identifier.
agency	factor	MCS LC	Added column for agency in charge.
city	factor	Springfield	City.
call_type	factor	Agitation	Call type or nature of incident.
call_outcome	factor	Remained in community	Call outcome.
dispatch_time	datetime	2025-12-17 03:14:00	Date and time of dispatch.
client_engagement_time	datetime	2025-12-17 03:38:00	Date and time of client engagement.
arrival_time	datetime	2025-12-17 03:35:00	Date and time of arrival.
clear_time	datetime	2025-12-17 03:48:00	Date and time of incident end.
dispatch_status	factor	Engaged	Status of dispatch (e.g., Engaged, Refused).
minutes_request_dispatch	double	17	Duration in minutes from call to dispatch.
minutes_dispatch_arrival	double	21	Duration in minutes from dispatch to arrival.
minutes_arrival_engagement	double	3	Duration in minutes from arrival to engagement.
minutes_arrival_departure	double	13	Total time on scene.

V Final data table

Variable	Type	Example Value	Description
<code>timestamp</code>	datetime	2015-01-01 00:00:00	Date and time.
<code>incident_id</code>	character	cad_15000001	Unique alphanumeric identifier.
<code>EPD</code>	factor	1	Binary indicator specifying whether the Eugene Police Department responded.
<code>SPD</code>	factor	0	Binary indicator specifying whether the Springfield Police Department responded.
<code>CAHOOTS</code>	factor	0	Binary indicator specifying whether the CAHOOTS mobile crisis unit responded.
<code>MCSLC</code>	factor	0	Binary indicator specifying whether the Mobile Crisis Services of Lane County responded.
<code>city</code>	factor	Eugene	City.
<code>nature</code>	factor	Person Stop	Initial recorded nature, problem type, or motive of the call.
<code>outcome</code>	factor	Arrest	Call outcome.
<code>dispatch_status</code>	factor	Engaged	Tracking marker specifying the dispatch categorization related to MCSLC operations (unavailable for other agencies).
<code>priority</code>	factor	6	Dispatch priority code assigned to the call, range 0-9 or P (unknown classification).
<code>source</code>	factor	CAD	Original administrative log system source where the data was extracted.

VI Parallel trends check

Table 4

	<i>Dependent variable:</i>
	arrest_rate
week_num	0.00002 (0.00003)
treated	0.019*** (0.002)
week_num:treated	0.00002 (0.00003)
Constant	0.009*** (0.002)
Observations	236
R ²	0.611
Adjusted R ²	0.606
Residual Std. Error	0.139 (df = 232)
F Statistic	121.234*** (df = 3; 232)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

VII Plots

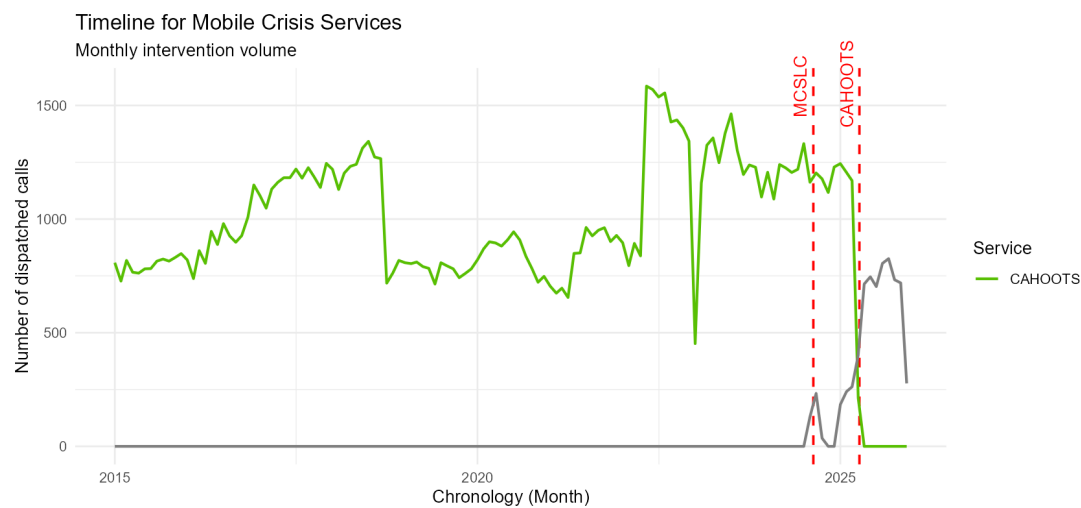


Figure 2: Call volume for MCR agencies in Eugene

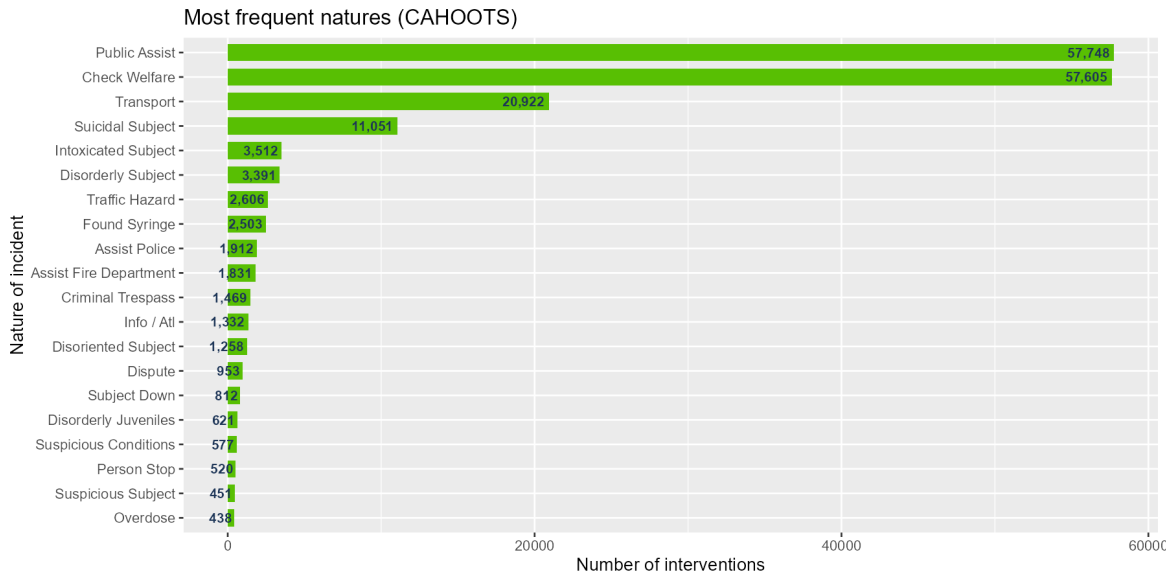


Figure 3: CAHOOTS all time most frequent call natures

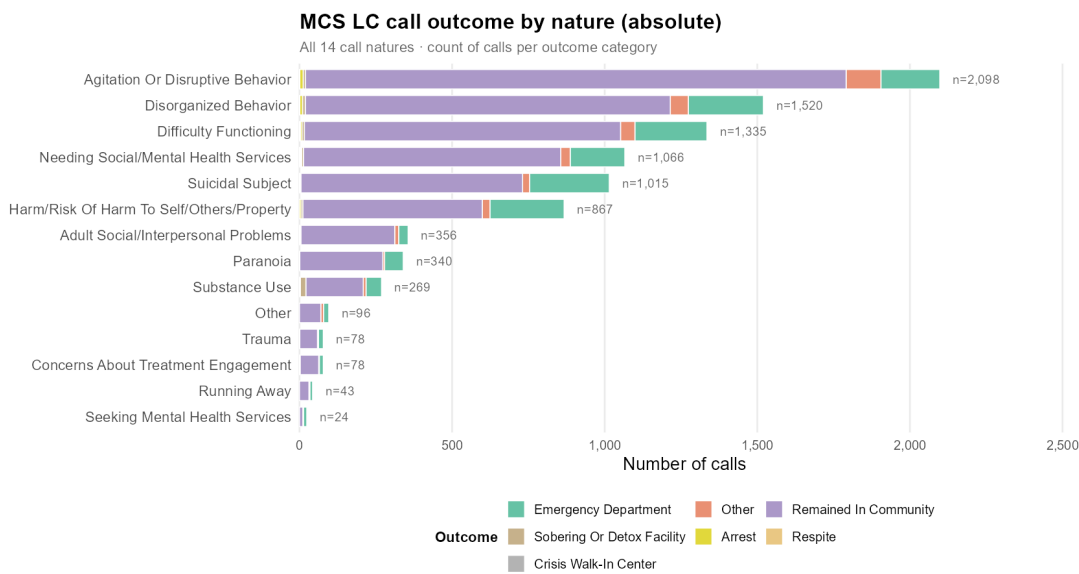


Figure 4: MCS-LC call outcome distribution per call nature

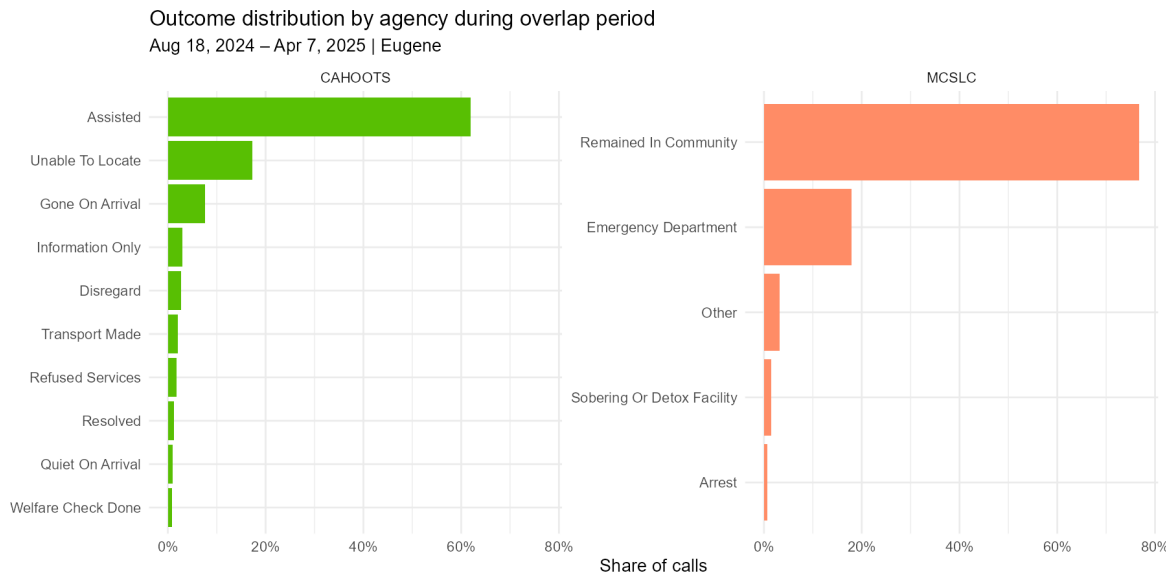


Figure 5: Outcome distribution by agency during overlap period

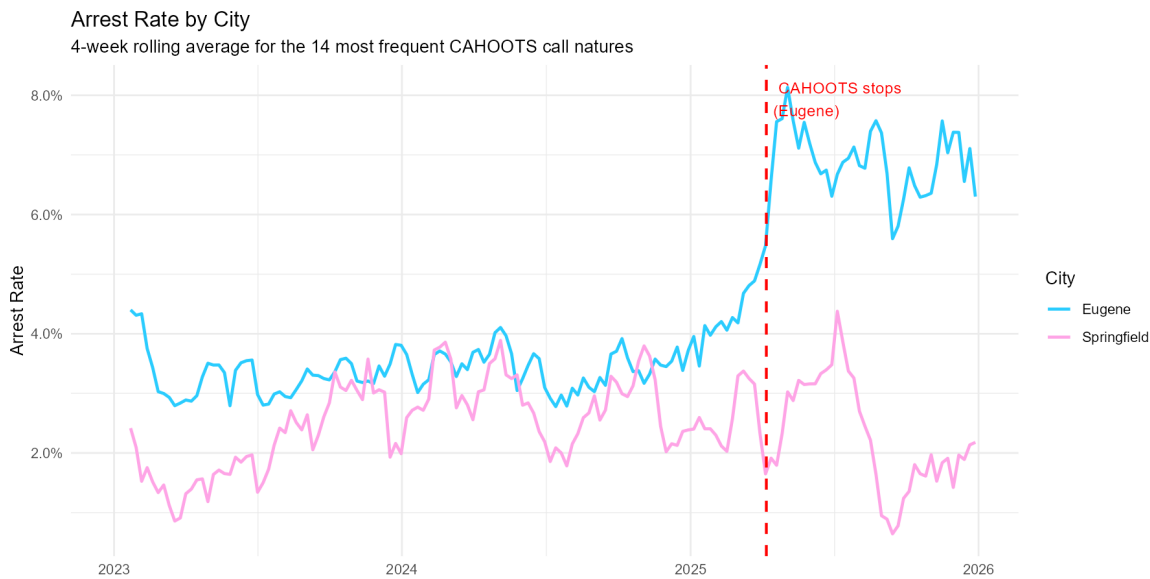


Figure 6: Arrest rate by city

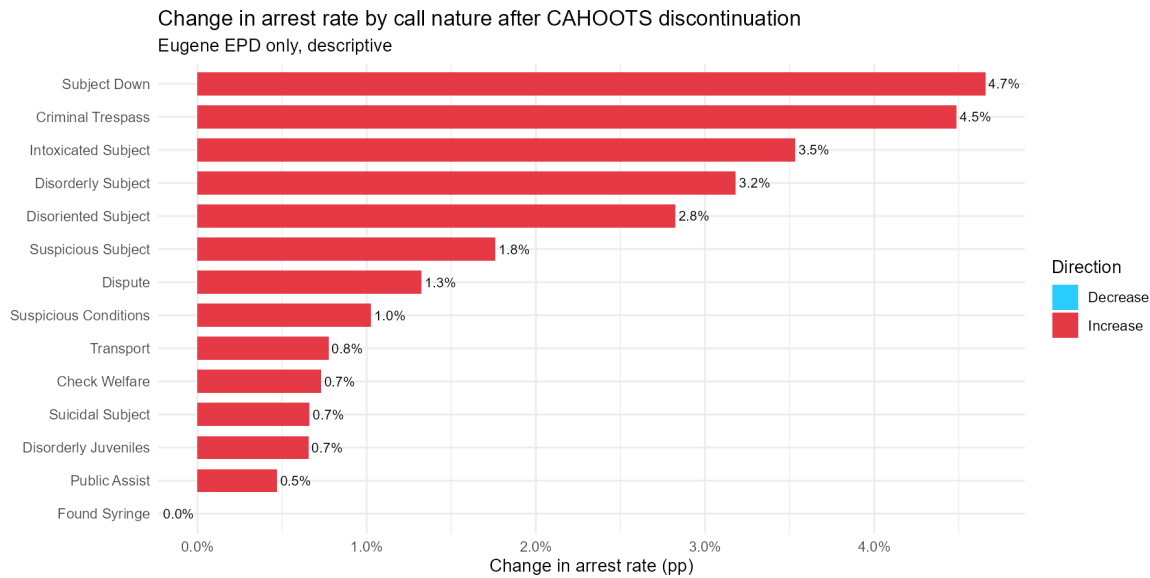


Figure 7: Arrest rate variation by call nature for EPD after CAHOOTS’ discontinuation

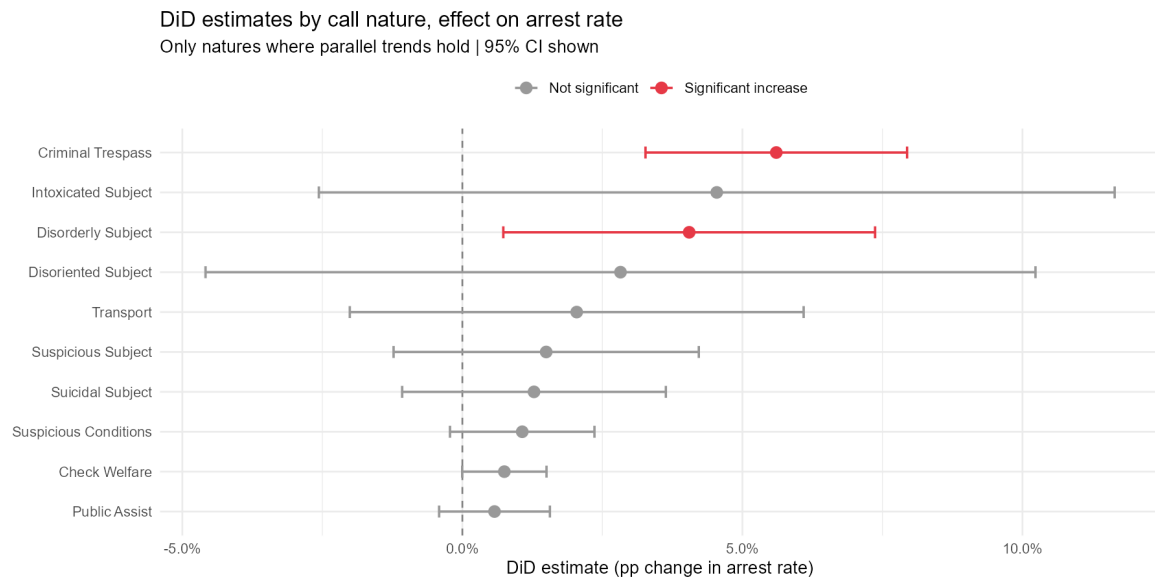


Figure 8: DiD results by call nature

Total nature-specific DiD estimates from pooled interaction model

nature	total_did	std.error	p.value	conf_low	conf_high
Criminal Trespass	0.0560	0.0092	0.0000	0.0380	0.0741
Subject Down	0.0466	0.0538	0.4681	-0.0590	0.1521
Intoxicated Subject	0.0454	0.0422	0.3689	-0.0373	0.1281
Disorderly Subject	0.0405	0.0118	0.0051	0.0174	0.0636
Disoriented Subject	0.0282	0.0622	0.7387	-0.0936	0.1501
Transport	0.0204	0.0297	0.6641	-0.0379	0.0787
Dispute	0.0168	0.0089	0.2968	-0.0007	0.0342
Suspicious Subject	0.0150	0.0145	0.6067	-0.0135	0.0434
Suicidal Subject	0.0128	0.0192	0.7831	-0.0249	0.0505
Suspicious Conditions	0.0107	0.0114	0.7788	-0.0116	0.0330
Disorderly Juveniles	0.0102	0.0233	0.9059	-0.0353	0.0558
Check Welfare	0.0075	0.0064	0.2451	-0.0051	0.0201
Public Assist	0.0057	0.0177	0.9210	-0.0290	0.0404
Found Syringe	0.0000	0.1378	0.9566	-0.2700	0.2700

Figure 9: Interaction model estimates

References

- [1] White Bird Clinic. What is CAHOOTS? <https://whitebirdclinic.org/what-is-cahoots/>. Accessed in 2026.
- [2] Lane County. What is MCS LC. https://www.lanecountyor.gov/government/county_departments/health_and_human_services/behavioral_health/mobile_crisis_services_of_lane_county/mobile_crisis_services_of_lane_county_f_a_qs. Accessed in 2026.
- [3] Rohlfslab. Rohlfslab website. <https://rohlfslab.weebly.com/>. Accessed in May 2026.
- [4] Big Ten Academic Alliance. International research experience program (irep), 2026. URL [https://btaa.org/international/international-research-experience-program-\(irep\)](https://btaa.org/international/international-research-experience-program-(irep)). Accessed in 2026.
- [5] Réseau FIGURE. Réseau figure cmi, 2026. URL <https://www.reseau-figure.fr/>. Accessed in 2026.
- [6] Wikipedia. Big ten conference wikipedia, 2026. URL https://en.wikipedia.org/wiki/Big_Ten_Conference. Accessed in 2026.
- [7] Wikipedia. Big ten conference wikipedia, 2026. URL https://en.wikipedia.org/wiki/Big_Ten_Academic_Alliance. Accessed in 2026.
- [8] SyNSec Lab. Synsec website. <https://synsec-den.github.io/>. Accessed in May 2026.
- [9] Wikipedia. University of oregon. https://en.wikipedia.org/wiki/University_of_Oregon. Accessed in 2026.
- [10] University of Oregon. About the university of oregon. <https://www.uoregon.edu/about>. Accessed in 2026.
- [11] IE². Ie² research. <https://ie2.uoregon.edu/research>. Accessed in May 2026.
- [12] University of Oregon School of Computer and Data Sciences. Scds. <https://scds.uoregon.edu/ds>, 2026. Accessed in 2026.

- [13] Carolina L. Adam, Joana Rocha, Peter Sudmant, and Rori Rohlf. Comparative genomics of tandem repeat variation in apes. *bioRxiv*, 2026. doi: 10.64898/2026.01.20.700717. URL <https://www.biorxiv.org/content/early/2026/01/21/2026.01.20.700717>.
- [14] Alicia Bierly and Melody Chang. Tandem repeats in adaptive archaic introgression across the human genome. Poster presented at the Undergraduate Research Symposium, University of Oregon, Accessed in 2026 2026.
- [15] Luis Fernandez-Luna, Sebastián Iturbe, Carolina Adam, Nathaniel S Pope, Diego Ortega-Del Vecchyo, and Rori Rohlf. A statistical framework to infer the mutation model of tandem repeat variants. *bioRxiv*, 2026. doi: 10.64898/2026.01.21.700917. URL <https://www.biorxiv.org/content/early/2026/01/22/2026.01.21.700917>.
- [16] N. Burton, C. Herbert, and R. Rohlf. Quantifying CAHOOTS: Mobile crisis response program diverts and prevents police response. Report, White Bird Clinic, 2024.
- [17] Jonathan Davis, Samuel Norris, Jadon Schmitt, Yotam Shem-Tov, and Chelsea Strickland. Mobile crisis response teams support better policing: Evidence from cahoots. Working Paper 33761, National Bureau of Economic Research, May 2025. URL <http://www.nber.org/papers/w33761>.
- [18] Nathan Wilk. Eugene after cahoots: The end of the crisis response program, and the efforts to bring it back. *KLCC*, June 2025. URL <https://www.klcc.org/health-medicine/2025-06-03/eugene-after-cahoots-the-end-of-the-crisis-response-program-and-the-efforts-to->
- [19] U.S. House of Representatives. H.r. 7961 – safety from violence for healthcare employees act. <https://www.congress.gov/bill/117th-congress/house-bill/7961>, 2022. 117th Congress, introduced June 7, 2022.
- [20] National Alliance on Mental Illness. Mobile crisis teams: Providing an alternative to law enforcement for mental health crises, July 2024. URL <https://www.nami.org/blog/mobile-crisis-teams-providing-an-alternative-to-law-enforcement-for-mental-heal> Accessed: June 22, 2026.
- [21] Eugene Police Department Crime Analysis Unit. Gap Analysis of Eugene’s Crisis Response Services Post- CAHOOTS Closure. Technical re-

- port, City of Eugene, 2021. URL <https://npr.brightspotcdn.com/a8/71/48aeabe1473bb1c8491a784ce90b/machine-readable-version.pdf>.
- [22] Mathilde Lafay. CAHOOTS: Data Analysis and Impact Evaluation of Alternative Emergency Response. <https://github.com/mathilde-ly/CAHOOTS>, 2026.
- [23] Hadley Wickham. Tidy data. *Journal of Statistical Software*, 59(10):1–23, 2014.
- [24] Law Enforcement Action Partnership. Eugene, Oregon CAHOOTS program. <https://lawenforcementactionpartnership.org/eugene-oregon-cahoots-program/>. Accessed in 2026.

VIII International experience

During the 2025-2026 academic year, I completed a semester-long academic exchange at Université Laval in Québec City, Canada. I enrolled in the Computer Science and Artificial Intelligence Master's program. The main language of instruction and in daily life was French. I took five classes, including Machine Learning, Artificial Intelligence, Database architecture, R Programming and Time series. This immersion into a North American campus environment was definitely a great experience, both academically and personally. Navigating the organizational and academic differences between the French and Canadian system, while discovering the Québec province and landscapes, has significantly strengthened my workload capacity, time management skills, and adaptability into an unfamiliar environment far from home.

For the Spring of 2026, I began my M1 internship through the International Research Experience Program (IREP) from the Big Ten Academic Alliance and the FIGURE network partnership. I spent 13 weeks as a graduate visiting researcher at the Rohlfslab, University of Oregon (USA).

The primary language used for all research activities and daily life was English and most of the professional interactions took place in person at the lab, in Eugene, Oregon. This highly instructive internship has provided me my first experience in a fully English-speaking environment as well as my first research experience. I learned a lot about the US Crisis Services Program and the local 911 and emergency services dispatch protocols here in Eugene. Conducting this research project focused on social impact has given me a better idea of how Data Science can be used for public interest, how to conduct a research project from start to finish and how to connect and collaborate with fellow researchers.

My overall experience at the Rohlfslab has been very positive. Along with the main research project, I had the opportunity to attend multiple research seminars within the Institute of Ecology and Evolution and the UO School of Data Science. These events, mostly focused on biology and computational methods, allowed me to discover new research fields, broaden my personal knowledge, and learn about different applied disciplines in Data Science. It has definitely helped me in defining my career goals and orienting my path towards academia and research in public or semi-public institutions, whether in France or abroad.